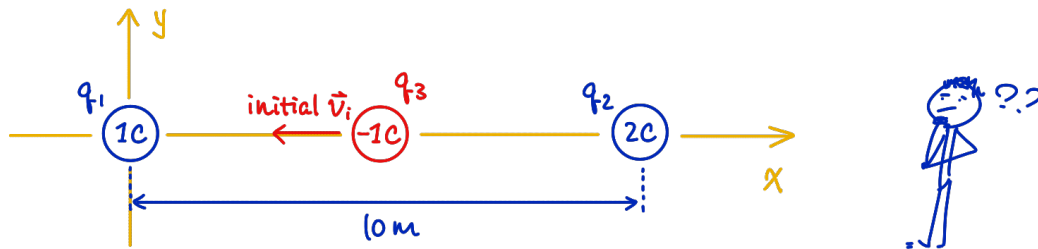


Physics 5C Discussion (Week 2)

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Problem 1: There are two point charges q_1 and q_2 fixed in space. Now Jinyan throws a third point charge q_3 into the region, such that q_3 will have some initial velocity, as indicated in the figure.



$q_1 = 1\text{ C}$ is fixed at $\vec{x}_1 = (0, 0)\text{ m}$.

$q_2 = 2\text{ C}$ is fixed at $\vec{x}_2 = (10, 0)\text{ m}$.

$q_3 = -1\text{ C}$ is initially between q_1 and q_2 on the x -axis with some initial velocity.

The Coulomb constant $k = 8.99 \cdot 10^9\text{ Nm}^2/\text{C}^2$.

Part (a) Sketch the electric field produced by the source charges q_1 and q_2 .

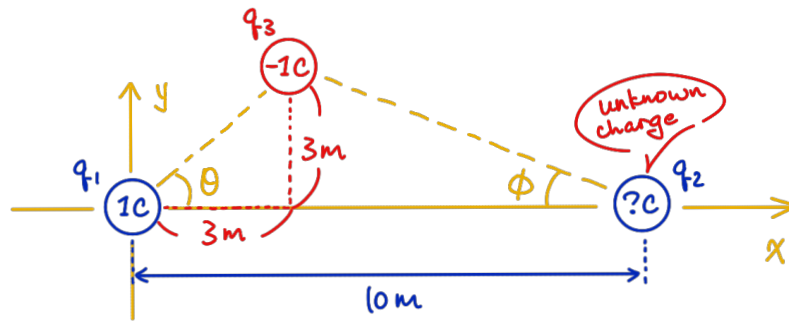
Part (b) At which point in space does the test charge q_3 experience a zero net force (neglect gravity and friction, only consider the electric force)?

Part (c) When q_3 is located at the point where it experiences zero net force, find the electric force experienced by q_2 in that configuration.

Part (d) Suppose q_3 is initially at $\vec{x}_3 = (5, 0)\text{ m}$, not the point where it experiences zero net force, with initial velocity $\vec{v}_i = (-1, 0)\text{ m/s}$. Qualitatively describe the motion of q_3 in such a configuration. How will it move under the electric forces from the source charges? Is there any way we can describe its motion quantitatively? ¹

¹No need to do any calculation for this part, just describe in words.

Problem 2: [This problem is not too difficult, try to solve it when you get time.]
Now we change the configuration a little bit, see the figure.

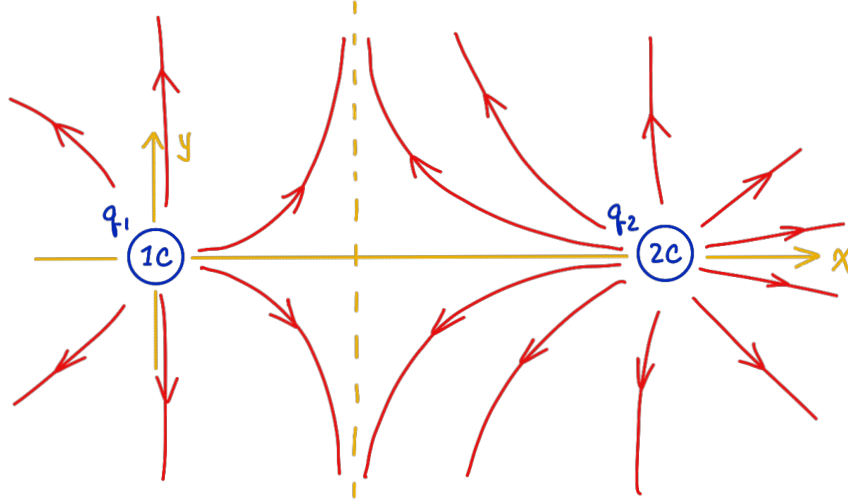


Now in this case, suppose we do not know the charge (or the sign) of q_2 fixed at $\vec{x}_2 = (10, 0)\text{ m}$, but we know that $q_1 = 1\text{ C}$ is fixed at $\vec{x}_1 = (0, 0)\text{ m}$ and $q_3 = -1\text{ C}$ placed at $\vec{x}_3 = (3, 3)\text{ m}$ has a tendency to move directly downwards (the electric force it experiences pulls it directly downwards). Find the electric field $\vec{E} = (E_x, E_y)$ at the position $\vec{x}_3 = (3, 3)\text{ m}$, due to the source charges q_1 and q_2 .²

²Hint: q_3 wants to move directly downwards, that means the horizontal components of the forces acting on q_3 sum up to zero. There are only two electric forces acting on q_3 , by the other two charges. Use some trigonometry to decompose the forces acting on q_3 , then find the charge of q_2 , and use that to find the electric field at $\vec{x}_3$ by superpositioning the electric fields due to q_1 and q_2 .

Problem 1

Field lines come out from positive charges and terminate at negative charges. Also note that $q_2 = 2\text{ C}$ is “stronger” than $q_1 = 1\text{ C}$.



Here q_3 experiences two forces, one due to q_1 and the other one due to q_2 . Since q_3 is negative, while q_1 and q_2 are positive, we deduce that both q_1 and q_2 are “pulling” q_3 . In particular, q_1 is dragging q_3 towards q_1 on the left, and q_2 is dragging q_3 towards q_2 on the right. Therefore, the location where q_3 experiences zero net force is the place where the two forces due to q_1 and q_2 cancel out exactly. It is not hard to see that this location must be somewhere between q_1 and q_2 on the horizontal x -axis.

To find such a location, let us suppose q_3 experiences zero net force when it is located at $(x, 0)$ with $0 < x < 10$ (because we want q_3 to be between q_1 and q_2). Then its distance to q_1 is x , and its distance to q_2 is $10 - x$, so we can write the two forces

$$\vec{F}_{13} = -\frac{k|q_1 q_3|}{x^2} \hat{x}, \quad \vec{F}_{23} = \frac{k|q_2 q_3|}{(10 - x)^2} \hat{x}.$$

The first one is negative because q_1 is dragging q_3 to the left. Both forces have their y -components vanish because q_1 and q_2 want to drag q_3 directly towards them. The net force q_3 experiences is thus the sum

$$\vec{F}_{\text{net}} = \vec{F}_{13} + \vec{F}_{23} = kq_3 \left(\frac{|q_2|}{(10 - x)^2} - \frac{|q_1|}{x^2} \right) \hat{x}.$$

We want to find where $\vec{F}_{\text{net}} = (0, 0)$, thus it suffices to find where

$$\frac{|q_2|}{(10 - x)^2} - \frac{|q_1|}{x^2} = 0.$$

Now we plug in the numbers,

$$\frac{2}{(10 - x)^2} - \frac{1}{x^2} = 0.$$

There are many ways to solve for x , and here is one of them,

$$\begin{aligned}\frac{2}{(10-x)^2} &= \frac{1}{x^2}, \\ 2x^2 &= (10-x)^2, \\ 2x^2 &= 100 - 20x + x^2.\end{aligned}$$

We obtain a quadratic equation, using the quadratic formula we obtain

$$x = 4.14 \text{ m}.$$

There is another solution $x = -24.14 \text{ m}$, but that one is not physical.

Now we want to find the electric force acting on q_2 , if we place q_3 at $\vec{x}_3 = (4.14, 0) \text{ m}$. In this case, since we want to find the force on q_2 , the “other” charges that are pulling or pushing q_2 are q_1 and q_3 . Therefore, q_2 is treated as a test charge, experiencing forces by the source charges q_1 and q_3 . Here q_1 and q_2 have the same sign, so they repel each other, and thus q_2 experiences a force to the right (positive sign) due to q_1 ,

$$\vec{F}_{12} = \frac{k|q_1 q_2|}{(\text{distance between } q_1 \text{ and } q_2)^2} \hat{x} = \left(\frac{8.99 \cdot 10^9 \cdot 1 \cdot 2}{10^2} \text{ N} \right) \hat{x} = (1.80 \cdot 10^8 \text{ N}) \hat{x}.$$

On the other hand, q_3 and q_2 have opposite signs, thus q_3 is pulling q_2 to the left (negative sign for the force). Now we write

$$\vec{F}_{32} = -\frac{k|q_3 q_2|}{(\text{distance between } q_3 \text{ and } q_2)^2} \hat{x} = \left(-\frac{8.99 \cdot 10^9 \cdot |-1| \cdot 2}{(10 - 4.14)^2} \text{ N} \right) \hat{x} = (-5.24 \cdot 10^8 \text{ N}) \hat{x}.$$

Adding these two forces together we get the net force experienced by q_2 ,

$$\vec{F}_{\text{net}} = \vec{F}_{12} + \vec{F}_{32} = (-3.44 \cdot 10^8 \text{ N}) \hat{x}.$$

After summing the effects due to both q_1 and q_3 , we see that q_2 is getting pulled to the left (hence the negative sign in the net force). This is because q_2 is closer to q_3 which pulls q_2 to the left.

In the problem, it is said that q_3 is initially at $\vec{x}_3 = (5, 0)$, to the right of the point where it experiences zero net force. Therefore, q_2 (on the right) would have a stronger effect on q_3 , trying to pull q_3 to the right. However, q_3 is initially moving to the left with $\vec{v}_i = (-1, 0) \text{ m/s}$, so q_3 will be moving to the left but slowing down. Depending on whether q_3 can pass through $x = 4.14 \text{ m}$ (the point where the forces due q_1 and q_2 balance out), there will be two different end results:

1. q_3 is initially moving to the left, slowing down, able to pass through $x = 4.14 \text{ m}$. In this case it enters the region where the force due to the left charge q_1 dominates, which pulls q_3 towards q_1 , so q_3 will end up hitting q_1 .
2. q_3 is initially moving to the left, slowing down, but unable to pass through $x = 4.14 \text{ m}$. Then in this case, it will “turn around” before reaching $x = 4.14 \text{ m}$, and remain in the region where the force due to the right charge q_2 dominates, which pulls q_3 towards q_2 , so q_3 will end up hitting q_2 .

Problem 2

In this configuration, we know that q_3 experiences a net force that has a zero x -component, because the net force is directly downwards. The force that q_1 exerts on q_3 has the form

$$\vec{F}_{13} = \left(-\frac{k|q_1q_3|}{r^2} \cos(\theta), -\frac{k|q_1q_3|}{r^2} \sin(\theta) \right),$$

where $r = \sqrt{3^2 + 3^2} \text{ m} = 3\sqrt{2} \text{ m}$ indicates the distance between charge q_1 and charge q_3 . The $\sin$ and $\cos$ are there to extract the x - and y -components of $\vec{F}_{13}$. We are able to do this (extract the components) because the magnitude of $\vec{F}_{13}$ is $|kq_1q_3|/r^2$ based on Coulomb's law. Similarly, we write down

$$\vec{F}_{23} = \left(\frac{k|q_2q_3|}{R^2} \cos(\phi), -\frac{k|q_2q_3|}{R^2} \sin(\phi) \right)$$

where $R = \sqrt{(10-3)^2 + 3^2} = \sqrt{58} \text{ m}$, and ϕ is the angle indicated on the original figure. The additional negative sign is added in the y -component to ensure that $\vec{F}_{23}$ is dragging q_3 towards q_2 (so we want the force to be pointing right-down, positive in x -component and negative in y -component).

Using the trigonometry here, we see that

$$\begin{aligned} \sin(\theta) &= \frac{3}{\sqrt{3^2 + 3^2}} = \frac{1}{\sqrt{2}}, & \cos(\theta) &= \frac{3}{\sqrt{3^2 + 3^2}} = \frac{1}{\sqrt{2}}. \\ \sin(\phi) &= \frac{3}{\sqrt{(10-3)^2 + 3^2}} = \frac{3}{\sqrt{58}}, & \cos(\phi) &= \frac{(10-3)}{\sqrt{(10-3)^2 + 3^2}} = \frac{7}{\sqrt{58}}. \end{aligned}$$

So we conclude

$$\vec{F}_{13} = \left(-\frac{k|q_1q_3|}{18} \frac{1}{\sqrt{2}}, -\frac{k|q_1q_3|}{18} \frac{1}{\sqrt{2}} \right), \quad \vec{F}_{23} = \left(\frac{k|q_2q_3|}{58} \frac{7}{\sqrt{58}}, -\frac{k|q_2q_3|}{58} \frac{3}{\sqrt{58}} \right)$$

The statement in the problem is that the sum of these two forces is directly downwards, that is the x -components should cancel each other,

$$-\frac{k|q_1q_3|}{18} \frac{1}{\sqrt{2}} + \frac{k|q_2q_3|}{58} \frac{7}{\sqrt{58}} = k|q_3| \left(-\frac{|q_1|}{18\sqrt{2}} + \frac{7|q_2|}{58\sqrt{58}} \right) = 0,$$

from which we can find q_2 by plugging in the value of q_1 ,

$$-\frac{1}{18\sqrt{2}} + \frac{7|q_2|}{58\sqrt{58}} = 0,$$

thus $q_2 = 2.48 \text{ C}$ (the sign of q_2 has to be positive, otherwise the x -components cannot cancel each other out based on the geometry, as both forces will be pushing q_3 to the left if q_2 was negative). Now the electric field at $\vec{x}_3 = (3, 3) \text{ m}$ is in fact the electric force experienced by q_3 divided by q_3 (note that q_3 is negative), that is,

$$\vec{E}(\vec{x}_3) = \frac{\vec{F}_{13} + \vec{F}_{23}}{q_3} = \left(\frac{k|q_1|}{18\sqrt{2}} - \frac{k|q_2|}{58} \frac{7}{\sqrt{58}}, \frac{k|q_1|}{18\sqrt{2}} + \frac{k|q_2|}{58} \frac{3}{\sqrt{58}} \right) \text{ N/C}.$$

Plug in $q_1 = 1 \text{ C}$ and $q_2 = 2.48 \text{ C}$ we find

$$\vec{E}(\vec{x}_3) = (0, 5.04 \cdot 10^8) \text{ N/C}.$$



A squirrel at the University of Michigan ... GO BLUE!